COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

■ Midterm Examination ☐ Terminal Examination – SPRING 2026

Course Title:	VLSI Design		Course Code:	CPE 434	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais		Programme	BCE		
Semester:		Batch:	Section:		Date:	
Time Allowed:	**90 Minutes**		**Maximum Marks:**		**25**	
Student's Name:			Reg. No.			

Important Instructions /Guidelines:

- Write your Student's Name and Reg. No. clearly on the first page.
- The exam is closed book, closed notes.
- Mobile phones are not allowed in the examination room.
- Sharing of Calculator is not allowed. (Use your own calculator).
- Attempt all questions.

Question 1:[CLO3-PLO3-C5] **Marks 9**

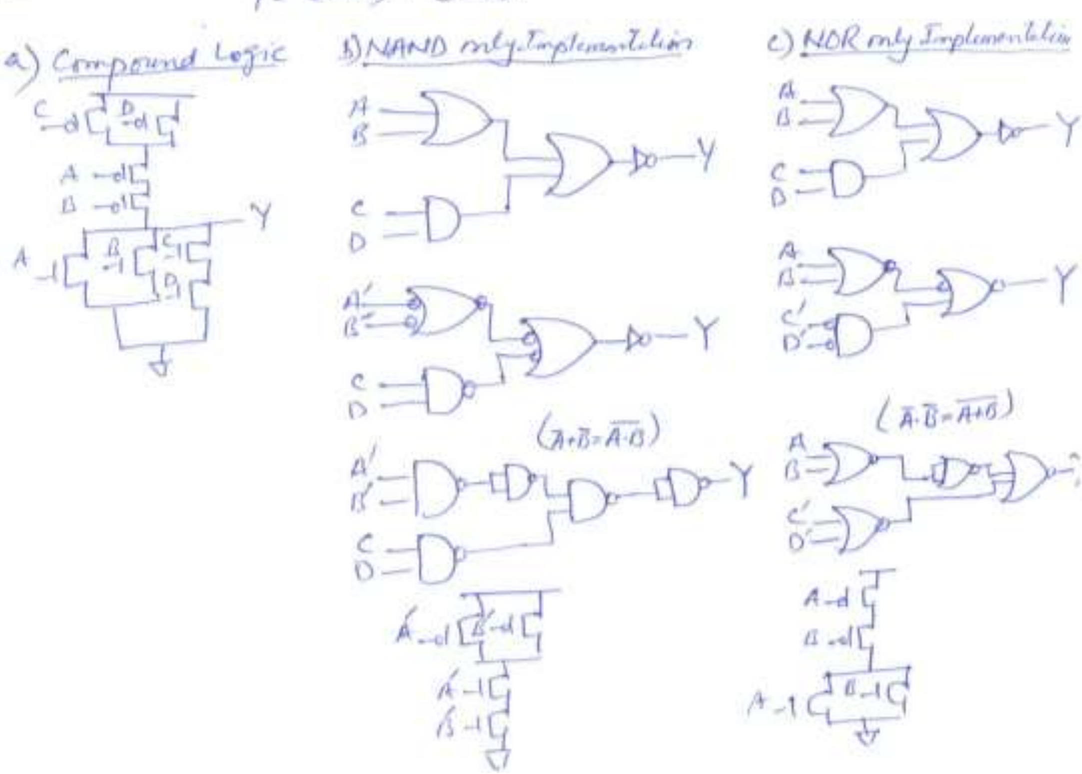
For the following Boolean function:

$$Y = \overline{(A+B) + (C.D)}$$

a) Design a CMOS circuit using compound logic
b) Design a CMOS circuit using NAND only logic (bubble pushing)
c) Design a CMOS circuit using NOR only logic (bubble pushing)

Question #1:-

$Y = \overline{(A+B) + (C\cdot D)}$



Question 2:[CLO1-PLO1-C3] **Marks 6**

Produce a stick diagram using *Euler's Path* for the following Boolean function:

$$X = \overline{A + (B.C.D) + (E.F)}$$

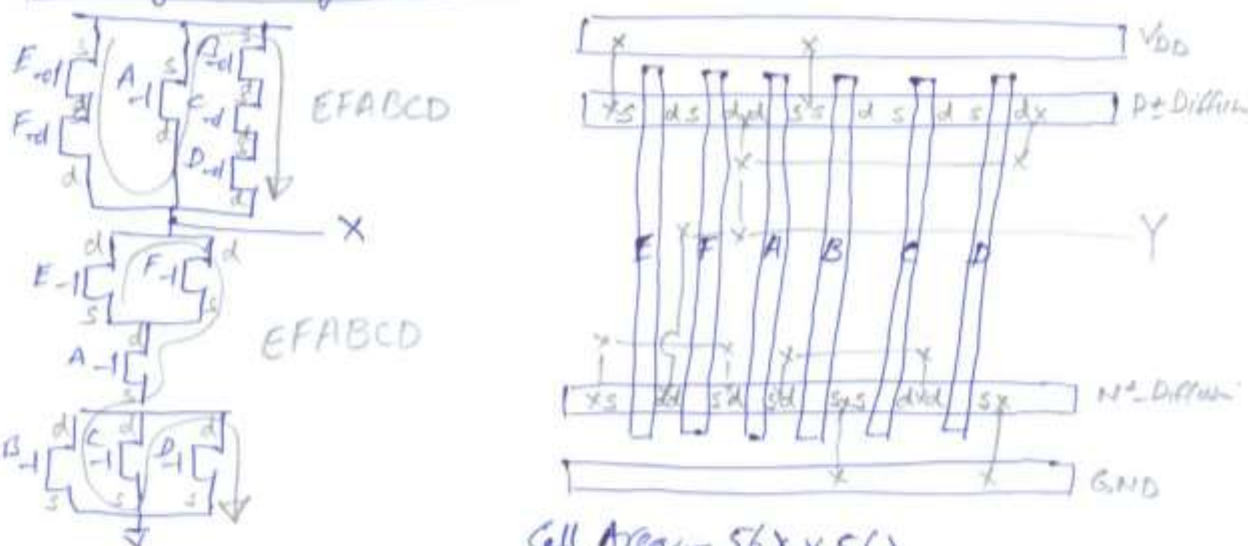
Analyze the stick diagram and calculate the area of a cell.

Question #2:-

$X = \overline{A+(B.C.D)+(E.F)}$

(Sample solution):- $X = \overline{A \cdot (B+C+D) \cdot (E+F)}$

Stick Diagram using Euler's Path:



Cell Area = 56λ x 56λ

Question 3:[CLO1-PLO1-C3] **Marks 10**

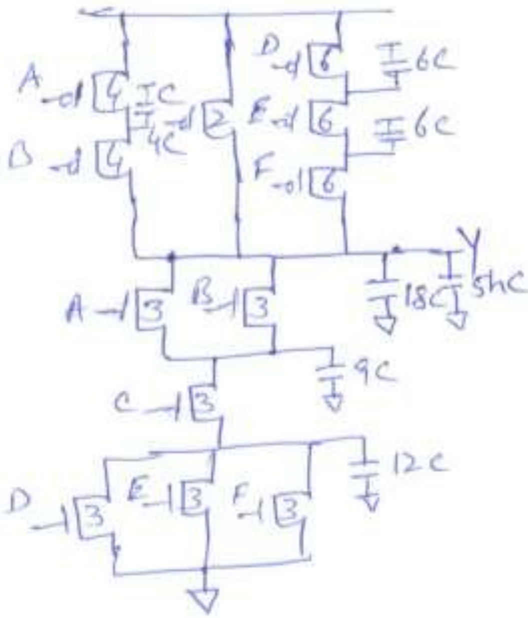
Apply an *RC Delay Model* to compute the delays for the following gate if the output is connected to the C input of the *h* identical gates:

$$Y = \overline{(A+B).C.(D+E+F)}$$

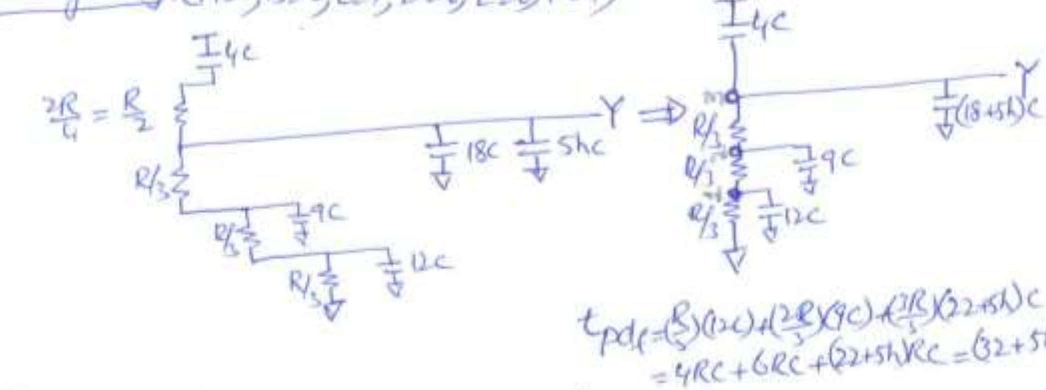
a) Falling propagation delay (t_{pdf}) for the input combination of A=1, B=0, C=1, D=0, E=0, and F=1
b) Rising propagation delay (t_{pdr}) for the input combination of A=0, B=0, C=1, D=1, E=1, and F=1

Question #3 :-

$$Y = \overline{(A+B) \cdot C \cdot (D+E+F)}$$

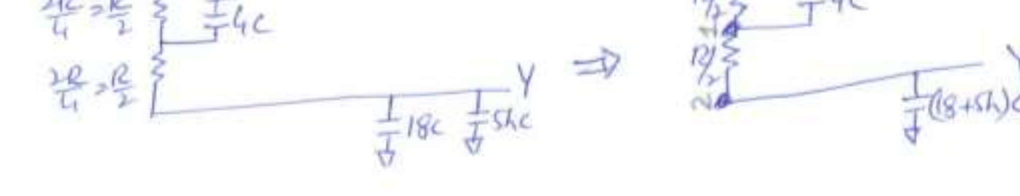


a) Falling Delay: $(A=1, B=0, C=1, D=0, E=0, F=1)$



$$t_{pdf} = \left(\frac{R}{3}\right)(12C) + \left(\frac{2R}{3}\right)(9C) + \left(\frac{3R}{3}\right)(22+5h)C$$
$$= 4RC + 6RC + (22+5h)RC = (32+5h)RC$$

b) Rising Delay $(A=0, B=0, C=1, D=1, E=1, F=1)$



$$t_{pdr} = \left(\frac{R}{2}\right)(4C) + \frac{2R}{2}(18+5h)C$$
$$= 2RC + (18+5h)RC = (20+5h)RC$$